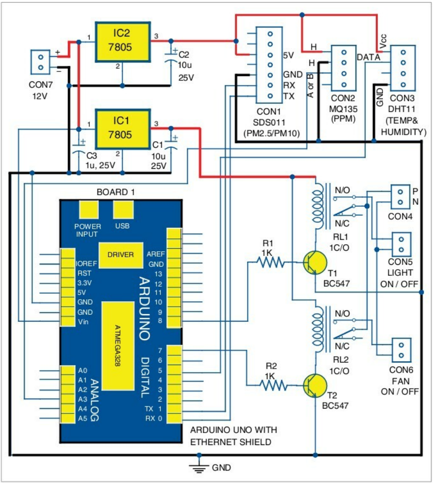

Fig. 3: Circuit diagram of the PM2.5/10 air pollution meter

of solid particles and liquid droplets floating in the air. Some particles are released directly from a specific source, while others are formed in complicated chemical reactions in the atmosphere.

The PM2.5/PM10 sensor connected across CON1 was developed by INOVAFIT, which is a spin-off from University of Jinan, China. It uses the principle of laser scattering in the air, and can detect suspended particulate matter concentration ranging from 0.3 to 10 microns. Data collected by the sensor is stable and reliable. SDS011 sensor is connected to UART port (T_x and R_x) of Arduino Uno board.

Gas sensor (MQ135). Sensitive material of the sensor is tin-dioxide, whose conductivity increases with the concentration of gas. Change in conductivity is converted into output voltage signal, which varies corresponding to the concentration of combustible gas. MQ135 is highly sensitive to ammonia, sulphide and benzene steams, smoke and other harmful gases. It is a low-cost sensor, suitable for different applications. Output of the gas sensor is connected to analogue input pin A3 of Arduino Uno board through connector CON2.

Temperature and humidity sensor (DHT11). This composite sensor contains calibrated digital signal outputs of temperature and humidity. Connected to connector CON3, it includes a resistive-type humidity measurement component and an NTC temperature-measurement device. Its output pin is connected to digital pin 5 of Arduino Uno board. It is a relatively inexpensive sensor for its performance.

Additional features. Additional alarm indication and control through relays RL1 and RL2 have been provided. The two appliances (light and fan) can be connected to connectors CON5 and CON6, but you can add more as per your requirement. Connector CON4 is used for connecting 230V AC mains to drive the lamp and fan connected at CON5 and CON6, respectively.

Power supply. A 12V battery or supply is connected to connector CON7, which is regulated to 5V using 7805 regulators (IC1 and IC2). When the system is powered on, the device takes one minute to preheat the gas sensor before it is ready for operation. The whole sensor circuit can be powered by 9V or 12V adaptor or battery. Here, IC1 and IC2 have been used for driving various parts of the circuit.

Enclose the whole sensor circuit (Fig. 3) in a suitable box, which should be placed where you want to monitor air quality or pollution.

Ethernet shield. Arduino Ethernet shield allows you to easily connect Arduino to the Internet. This shield enables Arduino to send and receive data from anywhere in the world, with an Internet connection.

Digital dashboard setup

Step 1. Mount Ethernet shield on Arduino Uno and connect Board1 to PC using a USB cable.

Step 2. Change the IP address in IPAddress IP (your IP address) in Arduino sketch ethernetclient.ino. This sketch/program code is included in this month's EFY DVD.

Compile and upload the code into Arduino Uno from Arduino IDE. This code shows you how to make an HTTP request using an Ethernet shield. It returns a Google search for the word Arduino. Results of this search are viewable as HTML through Arduino's serial window.

Step 3. Compile ethernetserver.